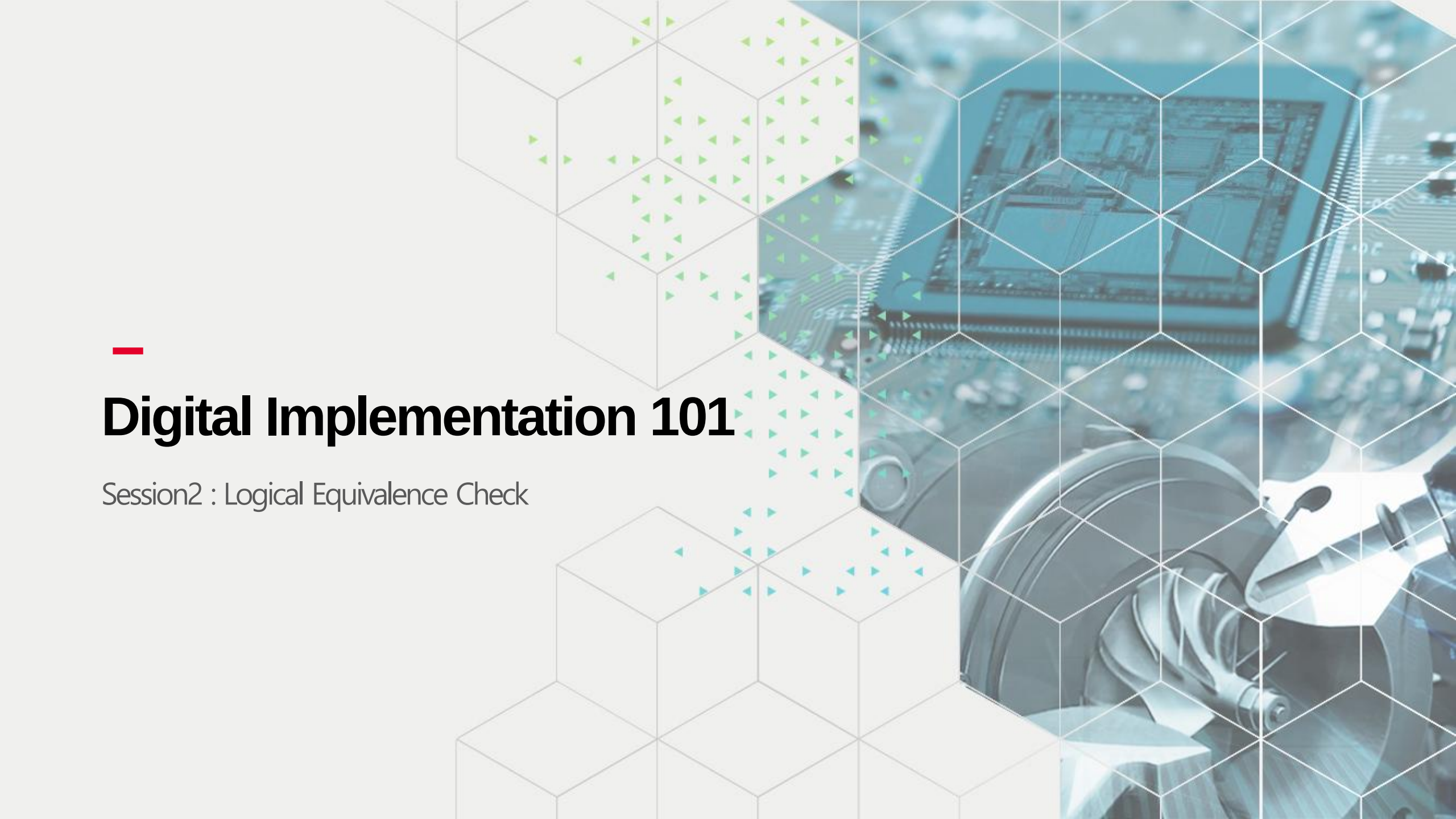


The background features a light gray hexagonal grid pattern. On the right side, there is a blue-tinted image of a circuit board with various components. Small green and blue triangles are scattered across the hexagonal grid, particularly concentrated in the upper and middle sections.

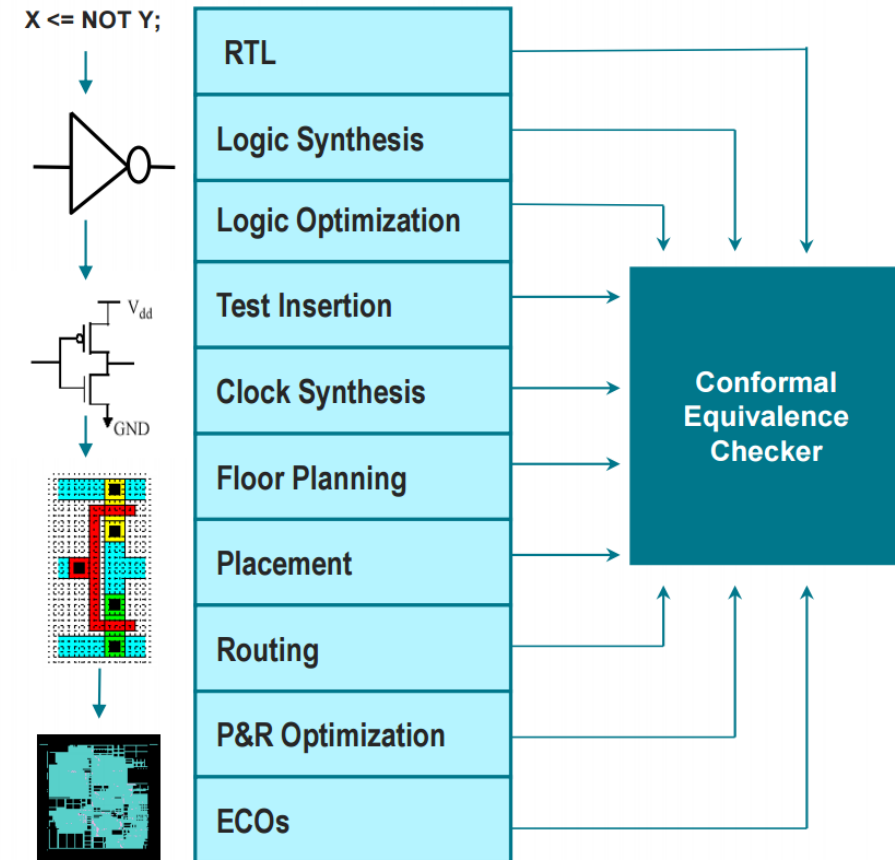
Digital Implementation 101

Session2 : Logical Equivalence Check

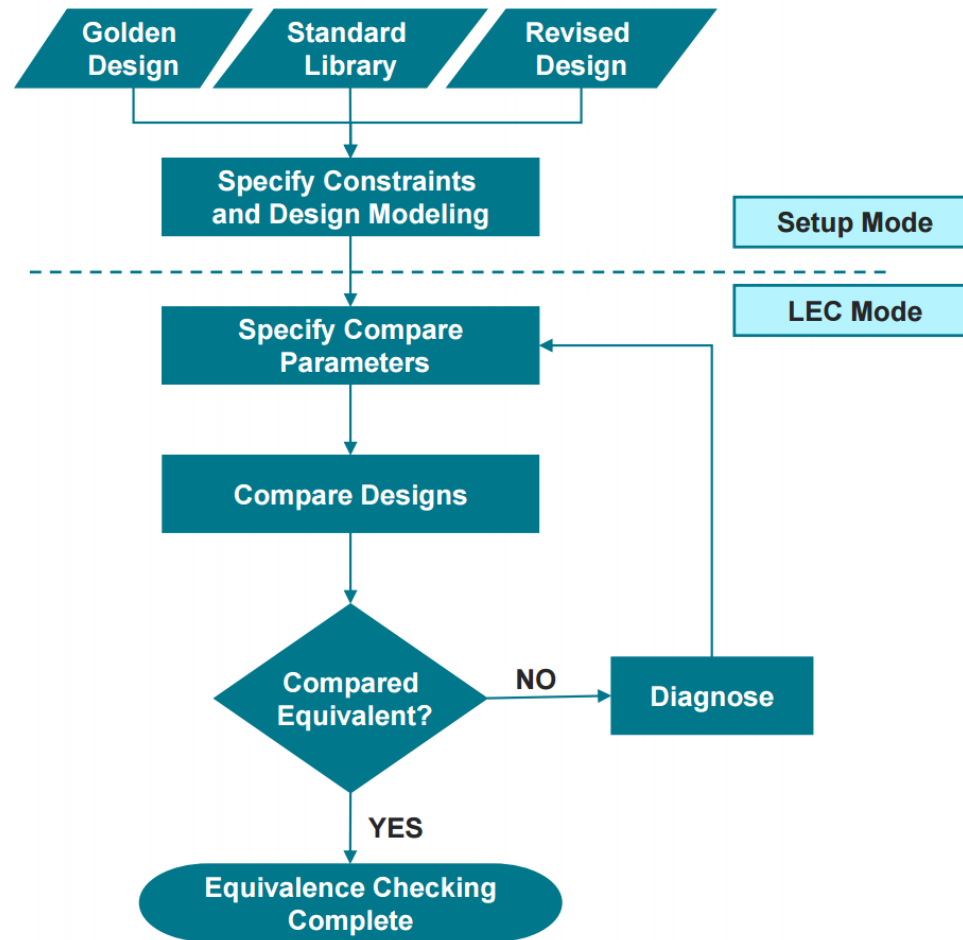
1. Understanding LEC

Notion of LEC

- RTL2gate and gate2gate equivalence checking
 - Verifies all transformations that occur during design implementation
 - Exhaustive verification without the use of test vectors
- Semantic and structural checks
 - Additional capabilities beyond EC to detect bugs earlier in the design cycle
- Safest EC solution
 - Independent verification- finds bugs that others miss
- Best comparison completion rate



LEC Flow



Setup Mode

- ✓ Specifying blackboxes
- ✓ Reading libraries and designs
- ✓ Specifying design constraints
- ✓ Specifying modeling directives

LEC Mode

- ✓ Mapping process
- ✓ Resolving unmapped key points
- ✓ Compare process
- ✓ Debugging nonequivalent key points

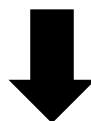
Input Files

LEC

PDK
(.lib/.db file for std/io cells)

RTL Source
(.v files for synthesis)

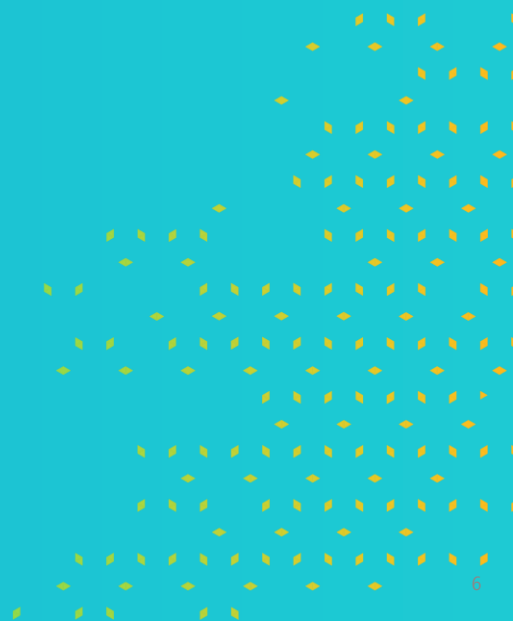
Gate Level Netlist



Equivalence Check
(Conformal)



LAB



실습

GUI

Gate Manager Mapping Manager Diagnosis Manager Find Module/Instance

Conformal(R) Logic Equivalence Checking

File Compare Tools Custom Preferences Window Help cadence

Design Library Tool Setup Design Data Rule Checking

Golden Revised

Design Hierarchy Window

Transcript Window

Command Status Bar

Modes

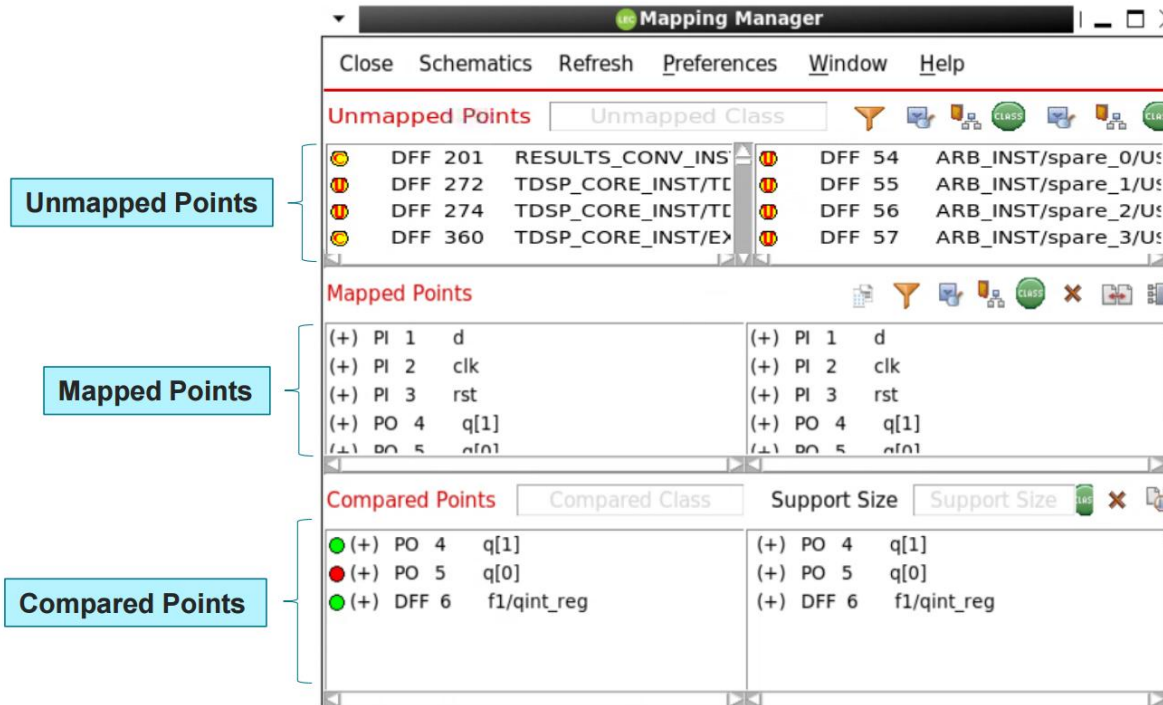
Status Bar

Golden	3	2	1	1	7
Revised	3	2	1	1	7
// 4 compared points added to compare list					
Compared points	PO	DFF	DLAT	Total	
Equivalent	2	1	1	4	

LEC> set system mode setup
SETUP> |

Compare done! 100% completed

Choose **Tools > Mapping Manager** to open the Mapping Manager, or click the mapping manager icon. 



Meanings of different colored circles in Unmapped Points:

Not-mapped 

Extra 

Unreachable 

Sequential Merge (S) 

Sequential Constant 

To show only unmapped points:

1. Choose **Preference > Sort by name**
2. Choose **Class > Disable All**
3. Choose **Class > Not-Mapped**

Mapping Manager – Comparison Results

Mapping Manager

CloseSchematicsRefreshPreferencesWindowHelp

Unmapped Points

Unmapped Class

DFF 12

const0

DFF 13

const1

DFF 14

c__0

DFF 15

c__1

Mapped Points

(+)

PI 1

set

(+)

PI 2

rst

(+)

PI 3

clk

(+)

PI 4

data

(+)

PI 5

enable

(+)

PO 6

x[5]

(+)

PO 7

x[4]

(+)

PI 1

set

(+)

PI 2

rst

(+)

PI 3

clk

(+)

PI 4

data

(+)

PI 5

enable

(+)

PO 6

x[5]

(+)

PO 7

x[4]

Compared Points

Compared Class

Support Size

Support Size

(+)

PO 6

x[5]

(+)

PO 7

x[4]

(+)

PO 8

x[3]

(+)

PO 9

x[2]

(+)

PO 10

x[1]

(+)

PO 11

x[0]

(+)

DFF 16

not_0

(+)

PO 6

x[5]

(+)

PO 7

x[4]

(+)

PO 8

x[3]

(+)

PO 9

x[2]

(+)

PO 10

x[1]

(+)

PO 11

x[0]

(+)

DFF 12

not_0

// Command: compare

Transcript window

Compared points	PO	DFF	Total
Equivalent	1	1	2
Non-equivalent	1	0	1

✓ Equivalent

✓ Inverted-Equivalent

✓ Sequential-Merge

✓ Non-Equivalent

✓ Abort

✓ Not-Compared

✓ Reachable

✓ Unreachable

✓ All

Disable All

- Equivalent
- Inverted-Equivalent
- Non-equivalent
- Abort
- Not-Compared
- Split status indicators show top-level sequential merge instances equivalence:
- Design - Equivalent

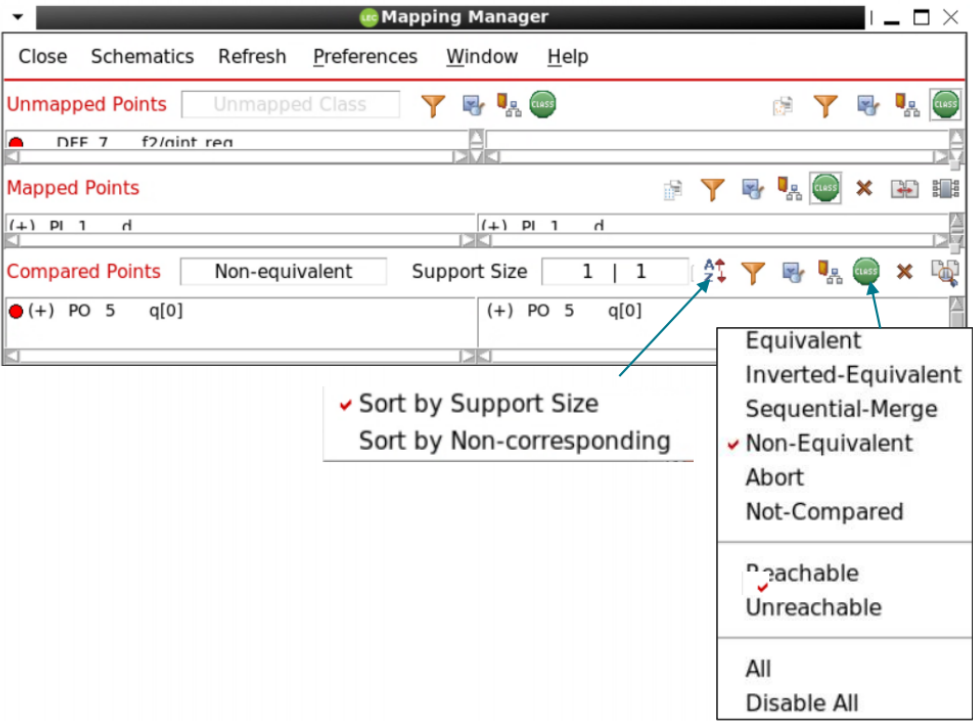
Sequential Merge - Equivalent
- Design - Equivalent

Sequential Merge - Abort

Mapping Manager – Comparison Results

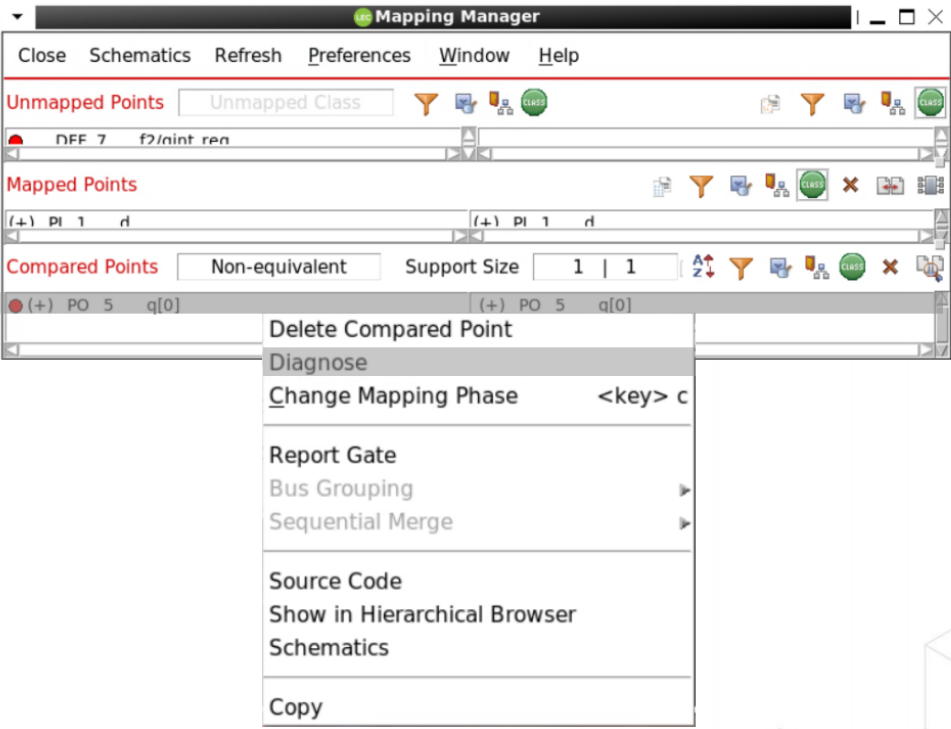
Use the Mapping Manager to display only nonequivalent results and sort them to show smaller cones first:

1. Choose Class > Disable All
2. Choose Class > Reachable and Non-Equivalent
3. Click the **AZ**↕ icon and select Sort by Support Size



After sorting the non-equivalent key points, open the Diagnosis Manager to further debug:

1. Left-click to select a nonequivalent point (red-filled circle)
2. Right-click and select Diagnose



Diagnosis Manager

Close Schematics Refresh Preferences Window Help

Golden		Revised	
Compared Point			
DFF 5 i0		DFF 5 i0	
Diagnosis Point (active)			
(1) PI 2 CLK [CLOCK]		(0) AND 14 i1	
Diagnosis Points (inputs)			
PI 2 CLK [CLOCK] MUX 10 U\$1 [DATA]		AND 14 i1 PI 3 DATA	
Corresponding Support			
(1) PI 2 CLK		(1) PI 2 CLK	
Non-corresponding Support			
		(0) DLAT 6 i2	
Error Pattern All 0s All 1s		Error Candidate	
1: 1 0 1 0		(1.00) AND 14 i1	

Annotations:

- Compared Points:** Points to the Compared Point section.
- Diagnosis Input Points:** Points to the Diagnosis Points (inputs) section.
- Corresponding Support:** Points to the Corresponding Support section.
- Non-corresponding Support:** Points to the Non-corresponding Support section.
- Error Pattern:** Points to the Error Pattern section.
- Active Diagnosis Input Points:** Points to the Diagnosis Point (active) section.
- Error Candidates:** Points to the Error Candidate section.

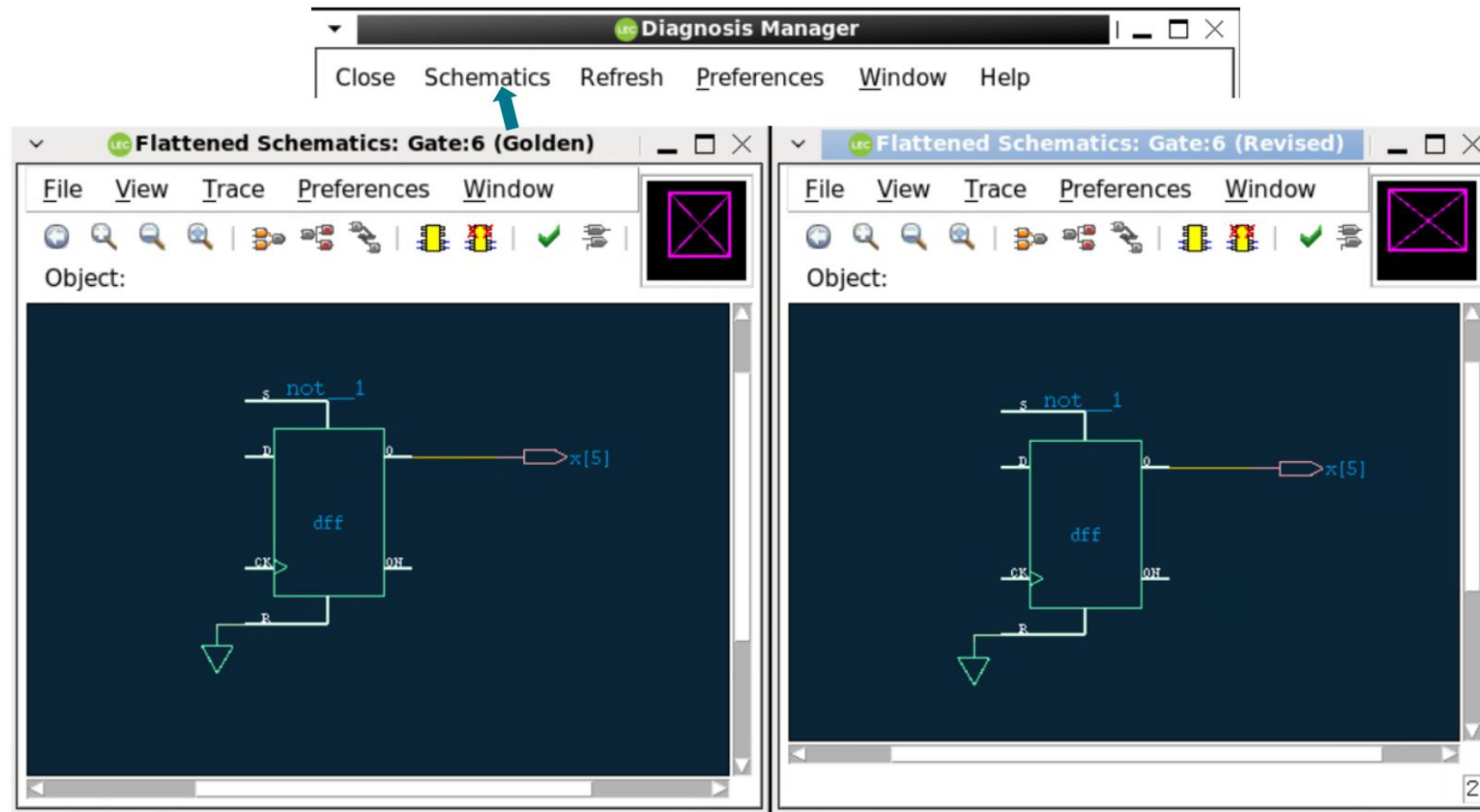
Legend:

- Non-corresponding and not mapped
- M Non-corresponding and mapped

Error Candidates: Points to possible root cause

Open schematic viewer from Diagnosis Manager:

- ✓ Right-click on any gate to open it's fanin or fanout cones.
- ✓ Double-click any gate to open Source Code.



From the Diagnosis Manager or the Schematic, you can open the Source Code Manager.

Signals can be traced across module boundaries.

The screenshot shows the Source Code Manager window with two panels. The left panel displays the source code for the 'golden' module, and the right panel displays the source code for the 'revised_1' module. Both panels show a history of changes and a search pattern field at the bottom.

```

1 //
2 //
3 module golden(OUT, ENABLE, CLK, DATA);
4 input ENABLE, CLK, DATA;
5 output OUT;
6
7 DFF i0(OUT, , 1'b0, 1'b0, CLK, ENABLE);
8 endmodule
    
```

```

1 //
2 module revised_1(OUT, ENABLE, CLK, DATA);
3 input ENABLE, CLK, DATA;
4 output OUT;
5
6 DFF i0(OUT, , 1'b0, 1'b0, ENCLK, DATA);
7 and i1(ENCLK, ENL, CLK);
8 DLAT i2(ENL, , 1'b0, 1'b0, ~CLK, ENABLE);
9 endmodule
10 //
11 module revised_2(OUT, ENABLE, CLK, DATA);
12 input ENABLE, CLK, DATA;
13 output OUT;
14
15 DFF i0(OUT, , 1'b0, 1'b0, ~ENCLK, DATA);
16 nand i1(ENCLK, ENL, CLK);
17 DLAT i2(ENL, , 1'b0, 1'b0, ~CLK, ENABLE);
18 endmodule
19 //
20 module revised_3(OUT, ENABLE, CLK, DATA);
21 input ENABLE, CLK, DATA;
    
```

do file : Automated Code from Genus

```
synthesis]$ lec -xl -color -dofile mapped/rtl_2_lec_final.do
```

```
=====  
Module Comparison Results  
-----  
Equivalent          3  
-----  
Total                3/3  
-----  
Hierarchical compare Equivalent  
-----  
0  
// Command: report_hier_compare_result -dynamicflattened  
// Command: report_hier_compare_result -dynamicflattened  
0  
// Command: report_verification -hier -verbose  
// Command: report_verification -hier -verbose
```

```
=====  
Verification Report  
-----  
Category                                                    Count  
-----  
1. Non-standard modeling options used:                      0  
   Tri-stated output:                                     checked  
   Revised X signals set to E:                             yes  
   Floating signals tied to Z:                             yes  
   Command "add clock" for clock-gating:                   not used  
-----  
2. Incomplete verification:                                  0  
   All primary outputs are mapped:                         yes  
   Not-mapped DFF/DLAT is detected:                         no  
   All mapped points are added as compare points:           yes  
   All compared points are compared:                         yes  
   User added black box:                                    no  
   Black box mapped with different module name:             no  
   Empty module is not black boxed:                         no  
   Command "add ignore outputs" used:                       no  
   Always false constraints detected:                        no  
-----  
3. User modification to design:                              0  
   Change gate type:                                       no  
   Change wire:                                           no  
   Primary input added by user:                             no  
-----  
4. Conformal Constraint Designer clock domain crossing checks recommended: 0  
   Multiple clocks in the design:                           no  
-----  
5. Design ambiguity:                                         1  
   Duplicate module definition:                             yes *  
   Black box due to undefined cells:                       no  
   Golden design has abnormal ratio of unreachable gates:   no  
   Ratio of golden unreachable gates:                       0%  
   Revised design has abnormal ratio of unreachable gates:   no  
   Ratio of revised unreachable gates:                       0%  
   All primary input bus ordering is consistent:             yes  
   All primary output bus ordering is consistent:            yes  
   DFF/DLAT not compared due to disabled clock port(s):      0  
   Always X compared point is detected:                     not checked  
-----  
6. Compare Results:                                          PASS  
   Number of EQ compare points:                             69  
   Number of NON-EQ compare points:                         0  
   Number of Aborted compare points:                        0  
   Number of Uncompared compare points :                    0  
=====
```


참고 – Abort의 예시

do file : RTL Source Code를 직접 지정 (no fv_map)

```
set log file ./log/all.log -replace
read library -verilog -both \
  ../../GPD045/digital/giolib045_v3.5/vlog/pads_SS_s1vg.v \
  ../../GPD045/digital/gsclib045_all_v4.4/gsclib045/verilog/slow_vdd1v0_basicCells.v
read design ../RTL/CORTEXM0DAP.v \
  ../RTL/CORTEXM0DS.v \
  ../RTL/cortexm0_wic.v \
  ../RTL/CORTEXM0INTEGRATION.v \
  ../RTL/cm0_dbg_reset_sync.v \
  ../RTL/cortexm0_rst_ctl.v \
  ../RTL/cortexm0ds_logic.v -verilog -golden
read design ../mapped/CORTEXM0INTEGRATION_syn_final.v -verilog -revised
set system mode lec
add compare points -all
compare
report verification
report statistics
```

Conformal(R) Logic Equivalence Checking

File Compare Tools Custom Preferences Window Help

Current Mode: LEC

Design Library Tool Setup Design Data Rule Checking Verification Report

Golden Revised

CORTEXM0INTEGRATION

- 1 library cells and 108 primitives
- u_cortexm0(CORTEXM0DS)
- u_dap(CORTEXM0DAP)
- u_dpreset_sync(cm0_dbg_reset_sync_PRESENT0)
- u_wic(cortexm0_wic)

CORTEXM0INTEGRATION

- 1 primitives
- u_cortexm0(CORTEXM0DS)

Memory usage : 211.34 M bytes

// Command: add compare points -all

// 1056 compared points added to compare list

// Command: compare

Compared points	PO	DFF	Total
Equivalent	134	598	732
Abort	0	324	324

CPU time : 812.24 seconds

Elapse time : 843 seconds

Memory usage : 294.09 M bytes

// Command: report verification

Verification Report

Category	Count
1. Non-standard modeling options used:	0
2. Incomplete verification:	0
3. User modification to design:	0
4. Conformal Constraint Designer clock domain crossing checks recommended:	0
5. Design ambiguity:	0
6. Compare Results:	ABORT

// Command: report statistics

Mapping and compare statistics

Compare Result	Golden	Revised
SETUP> dofile do.do		
LEC> diagnose 588 -golden // gate u_cortexm0/u_logic/A6ul85_reg		
LEC>		

Compare done!

100% completed

Diagnosis Manager

Close Schematics Refresh Preferences Window Help

Golden Revised

Compared Point

DFF 588 u_cortexm0/u_logic/A6ul85_reg DFF 289 u_cortexm0/u_logic/A6ul85_reg/U\$1/U\$1

Diagnosis Point (active)

(0) PI 3 [CLOCK] HCLK (0) INV 8689 u_cortexm0/u_logic/A6ul85_reg/U\$3

Diagnosis Points (inputs)

PI 3 [CLOCK] HCLK INV 8689 u_cortexm0/u_logic/A6ul85_reg/U\$3

MUX 21707 [DATA] u_cortexm0/u_logic/U\$15881 OR 20843 u_cortexm0/u_logic/q261467/U\$4

Corresponding Support

(0) PI 3 HCLK (0) PI 3 HCLK

Non-corresponding Support

Error Pattern All 0s All 1s Error Candidate

1:

Golden	Revised
(+) DFF 589 u_cortexm0/u_logic/L4ul85_reg	(+) DFF 663 u_cortexm0/u_logic/L4ul85_reg/U\$1/U\$1
(+) DFF 590 u_cortexm0/u_logic/W2ul85_reg	(+) DFF 1035 u_cortexm0/u_logic/W2ul85_reg/U\$1/U\$1
(+) DFF 591 u_cortexm0/u_logic/H1ul85_reg	(+) DFF 521 u_cortexm0/u_logic/H1ul85_reg/U\$1/U\$1
(+) DFF 592 u_cortexm0/u_logic/Sztl85_reg	(+) DFF 927 u_cortexm0/u_logic/Sztl85_reg/U\$1/U\$1
(+) DFF 593 u_cortexm0/u_logic/Dytl85_reg	(+) DFF 415 u_cortexm0/u_logic/Dytl85_reg/U\$1/U\$1

Flattened Schematics: Gate:3 (Golden/Diagnosed)

File View Trace Preferences Window

Object:

Design Find Options

Golden

CORTEXM0INT

Flattened Schematics: Gate:8689 (Revised/Diagnosed)

File View Trace Preferences Window

Object:

Design Find Options

Revised

CORTEXM0INT